FINITE TOTAL COHOMOLOGY DIMENSION EQUALS COMPACTNESS FOR THE ENDOMORPHISM RING

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Allowed standard prerequisites (editorial).

- Compact objects of D(R). dga-lemma-compact (cross-chapter, named). Compact implies finite projective graded representation.
- Truncation triangles. derived-remark-truncation-distinguished-triangle, derived-lemma-compact-objects-subcategory (cross-chapter, named). Reduces to a single cohomology module.

Source reference. Original reference 09QH: focused-seeds/geometry/S9-eqv-perfect-for-R/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08J5: focused-seeds/geometry/S9-eqv-perfect-for-R/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09RB: focused-seeds/geometry/S9-eqv-perfect-for-R/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09R3: focused-seeds/geometry/S9-eqv-perfect-for-R/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With k, n, and R as above, for an object K of $D(R)$ the following are equivalent*

(1) $\sum_{i \in \mathbf{Z}} \dim_k H^i(K) < \infty$, *and*
(2) K *is a compact object.*

Proof. If K is a compact object, then K can be represented by a complex $M^\bullet$ which is finite projective as a graded R-module, see Differential Graded Algebra, Lemma 09RB. Since $\dim_k R < \infty$ we conclude $\sum \dim_k M^i < \infty$ and a fortiori

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$\sum \dim_k H^i(M^\bullet) < \infty$. (One can also easily deduce this implication from the easier Differential Graded Algebra, Proposition 09R3.)

Assume K satisfies (1). Consider the distinguished triangle of trunctions $\tau_{\leq m}K \to K \to \tau_{\geq m+1}K$, see Derived Categories, Remark 08J5. It is clear that both $\tau_{\leq m}K$ and $\tau_{\geq m+1}K$ satisfy (1). If we can show both are compact, then so is K, see Derived Categories, Lemma 09QH. Hence, arguing on the number of nonzero cohomology modules of K we may assume $H^i(K)$ is nonzero only for one i. Shifting, we may assume K is given by the complex consisting of a single finite dimensional R-module M sitting in degree 0.

Since $\dim_k(M) < \infty$ we see that M is Artinian as an R-module. Thus it suffices to show that every simple R-module represents a compact object of $D(R)$. Observe that

$$I = \begin{pmatrix} 0 & S_1 & S_2 & \ldots & \ldots \\ 0 & 0 & S_1 & \ldots & \ldots \\ 0 & 0 & 0 & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & \ldots & \ldots & \ldots & 0 \end{pmatrix}$$

is a nilpotent two sided ideal of R and that R/I is a commutative k-algebra isomorphic to a product of $n+1$ copies of k (placed along the diagonal in the matrix, i.e., R/I can be lifted to a k-subalgebra of R). It follows that R has exactly $n+1$ isomorphism classes of simple modules $M_0, \ldots, M_n$ (sitting along the diagonal). Consider the right R-module P_i of row vectors

$$P_i = \begin{pmatrix} 0 & \ldots & 0 & S_0 & \ldots & S_{i-1} & S_i \end{pmatrix}$$

with obvious multiplication $P_i \times R \to P_i$. Then we see that $R \cong P_0 \oplus \ldots \oplus P_n$ as a right R-module. Since clearly R is a compact object of $D(R)$, we conclude each P_i is a compact object of $D(R)$. (We of course also conclude each P_i is projective as an R-module, but this isn't what we have to show in this proof.) Clearly, $P_0 = M_0$ is the first of our simple R-modules. For P_1 we have a short exact sequence

$$0 \to P_0^{\oplus n+1} \to P_1 \to M_1 \to 0$$

which proves that M_1 fits into a distinguished triangle whose other members are compact objects and hence M_1 is a compact object of $D(R)$. More generally, there exists a short exact sequence

$$0 \to C_i \to P_i \to M_i \to 0$$

where C_i is a finite dimensional R-module whose simple constituents are isomorphic to M_j for $j < i$. By induction, we first conclude that C_i determines a compact object of $D(R)$ whereupon we conclude that M_i does too as desired. $\square$